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# Decentralized Weapon-Target Assignment in Aerospace Defense Systems: An Attention-Enhanced Multi-Agent Reinforcement Learning Approach

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**Abstract**

Air defense weapon-target assignment (WTA) is a fundamental decision-making problem in aerospace defense systems, where decentralized coordination among heterogeneous radar and missile-launcher units is critical for effective air-defense operations. However, dynamic battlefield evolution, complex spatiotemporal interactions among heterogeneous assets and targets, and conflicting operational objectives under uncertain engagement outcomes pose significant challenges to existing multi-agent reinforcement learning methods. To address these challenges, this paper proposes a spatiotemporal graph attention-enhanced multi-objective advantage decomposition actor-critic (STG-MADAC) framework for decentralized WTA. Specifically, a spatiotemporal graph attention mechanism captures the evolving interactions among radars, missile launchers, and targets to provide structured situational representations, while a multi-objective advantage decomposition mechanism adaptively balances engagement effectiveness, resource consumption, and residual threat during policy optimization. A sequential heterogeneous actor-update strategy is further introduced to improve coordination among agents with different operational roles. Simulation results demonstrate that STG-MADAC consistently achieves superior overall performance compared

with representative multi-agent reinforcement learning methods across multiple mission-level metrics. Interpretability analyses further show that the proposed framework effectively captures dynamic threat relationships and supports more coordinated engagement decisions.

**Keywords:** Air defense systems, weapon-target assignment, decentralized decision making, graph attention networks, multi-agent reinforcement learning

# 1 Introduction

Weapon-Target Assignment (WTA) is a fundamental decision-making problem in military operations research, originally formulated by Alan S. Mann [1]. In modern Ground-Based Air Defense (GBAD) systems, WTA aims to allocate limited defensive resources, including radar guidance systems, missile launchers, and interceptor missiles, to multiple incoming aerial threats in a timely and coordinated manner [2], [3]. As modern air-defense operations evolve toward decentralized command-and-control architectures, heterogeneous radar and missile-launcher units are required to cooperate autonomously under incomplete battlefield information and stringent time constraints. Consequently, effective decentralized WTA has become increasingly important for improving engagement effectiveness, resource utilization, and overall system survivability. However, the NP-complete nature of WTA, coupled with the dynamic and uncertain characteristics of realistic combat environments, makes it one of the most challenging decision-making problems in intelligent air-defense systems.

Despite recent advances in intelligent decision-making, several fundamental challenges remain unresolved in decentralized GBAD-WTA. First, battlefield situations evolve continuously over time, and the interactions among radar units, missile launchers, and maneuvering targets exhibit complex spatio-temporal dependencies that are difficult to represent using conventional state representations. Second, heterogeneous defense assets possess distinct sensing capabilities, operational functions, and decision spaces, making effective coordination under decentralized execution considerably more difficult than homogeneous multi-agent cooperation. Third, practical GBAD-WTA requires balancing multiple conflicting operational objectives, including interception effectiveness, resource consumption, and residual threat, under uncertain engagement outcomes. Existing learning-based methods often optimize these objectives through scalarized reward functions, which implicitly couple different objectives and limit the flexibility of policy optimization. These challenges jointly restrict the applicability of existing intelligent decision-making methods in dynamic air-defense scenarios.

Over the past decades, numerous approaches have been proposed to address the WTA problem. Early studies mainly relied on exact optimization techniques, such as integer programming [5] and dynamic programming [6], which can obtain globally optimal solutions for small-scale static scenarios but suffer from prohibitive computational complexity as problem size increases. To improve computational efficiency, various heuristic and meta-heuristic algorithms were subsequently developed. However, these methods generally lack the capability to exploit historical decision information

and adapt to continuously changing battlefield environments, limiting their effectiveness for dynamic WTA problems. More recently, deep reinforcement learning (DRL) has demonstrated promising capabilities for sequential decision-making in dynamic environments through adaptive policy optimization. In particular, multi-agent reinforcement learning (MARL) [11], [12], typically formulated under the decentralized partially observable Markov decision process (Dec-POMDP) framework [16], has provided an effective solution paradigm based on centralized training and decentralized execution (CTDE) [13]?C[15]. Nevertheless, existing MARL-based WTA methods still exhibit several limitations. Most studies rely on simplified battlefield representations that inadequately capture the dynamic interactions among heterogeneous battlefield entities. Although graph neural networks (GNNs) [29] and Transformer-based architectures [30] have recently been introduced for situation modeling, existing methods rarely characterize the evolving interactions among radar units, missile launchers, and targets within a unified spatio-temporal representation. Furthermore, most multi-objective reinforcement learning approaches employ weighted reward aggregation, making it difficult to independently regulate different operational objectives and leading to suboptimal policy learning under conflicting mission requirements.

Motivated by the above observations, this paper proposes a Spatio-Temporal Graph Attention-enhanced Multi-objective Advantage Decomposition Actor-Critic (STG-MADAC) framework for decentralized GBAD-WTA. The proposed framework is based on the observation that battlefield entities naturally constitute a dynamic heterogeneous interaction graph, while different operational objectives require differentiated optimization signals during policy learning. Specifically, a spatio-temporal graph attention mechanism is developed to capture the evolving interactions among radar units, missile launchers, and targets, providing structured battlefield representations for cooperative decision-making. A multi-objective advantage decomposition mechanism is introduced to decouple the optimization process of multiple operational objectives by adaptively regulating policy-update signals. In addition, a sequential heterogeneous actor-update strategy is designed to improve coordination among heterogeneous agents with different operational functions under decentralized execution. By integrating these components into a unified actor-critic architecture, the proposed framework enables more effective cooperative decision-making for dynamic GBAD-WTA problems.

The main contributions of this paper are summarized as follows:

*A heterogeneous multi-agent reinforcement learning framework for decentralized GBAD-WTA.* We formulate decentralized GBAD-WTA as a heterogeneous multi-agent decision-making problem and propose the STG-MADAC framework, which explicitly models cooperation among heterogeneous radar and missile-launcher units within a unified actor-critic architecture.

*A spatio-temporal representation and multi-objective policy optimization mechanism.* We develop a spatio-temporal graph attention encoder to model the evolving interactions among heterogeneous battlefield entities and propose a multi-objective advantage decomposition mechanism that enables adaptive optimization of multiple conflicting operational objectives during policy learning.

*An interpretable learning framework for decentralized air-defense decision-making.* We further establish an explainability analysis framework that associates learned attention distributions with cooperative engagement behaviors, providing insights into dynamic target prioritization and improving the transparency of deep reinforcement learning-based WTA decisions.

The remainder of this paper is organized as follows. Section 2 reviews related work on ground-based air defense and dynamic WTA. Section 3 presents the mathematical formulation of the GBAD-WTA problem. Section 4 describes the proposed STG-MADAC framework. Section 5 reports experimental results and analysis. Finally, Section 6 concludes the paper.

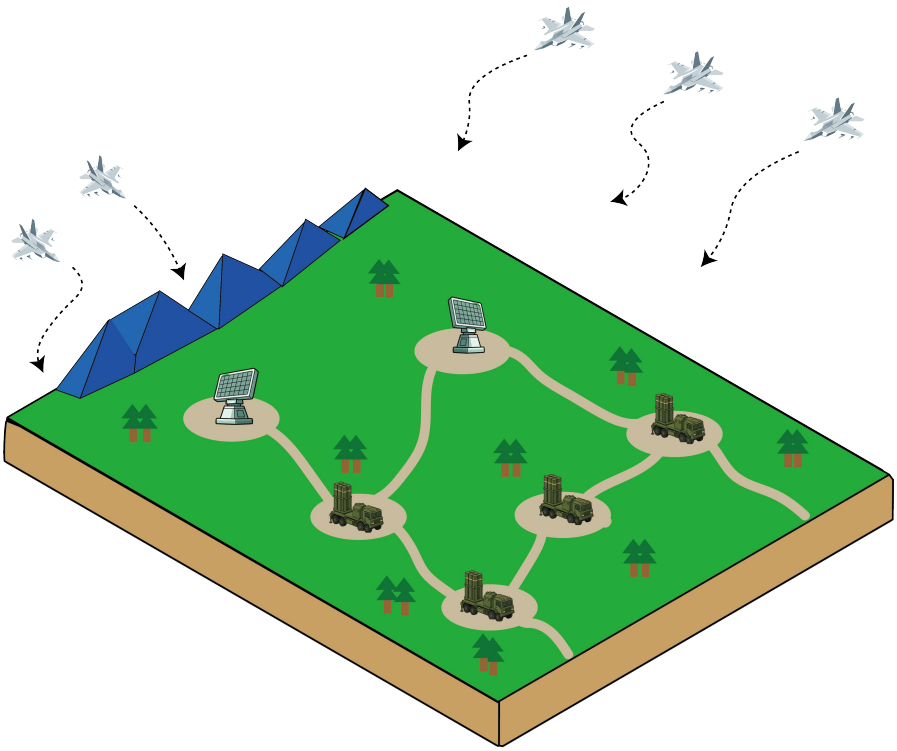


**Fig. 1** Ground-based air defense systems and situation display

## 2 Related work

Recent advances in weapon-target assignment have significantly improved decision-making capabilities in dynamic combat environments. Existing studies can generally be categorized into two research directions: optimization-based WTA methods and learning-based WTA methods. The former focuses on improving solution quality

through mathematical optimization and heuristic search, while the latter aims to enhance adaptability through data-driven sequential decision-making. Nevertheless, both research directions still face limitations in modeling heterogeneous battlefield interactions and optimizing multiple conflicting operational objectives.

## 2.1 Optimization-based WTA methods

Early WTA research mainly relied on exact optimization algorithms to obtain globally optimal solutions. Chang et al. [32] proposed an exact solution framework based on linear relaxation and branch-and-bound techniques, while Kline et al. [33] developed a Branch-and-Adjust algorithm using piecewise linear approximation, extending exact optimization to larger-scale WTA instances. Despite their optimality, the computational complexity of exact methods increases exponentially with problem size, making them unsuitable for real-time battlefield decision-making.

To improve computational efficiency, various heuristic and meta-heuristic algorithms were subsequently developed. Hocaoglu [35] introduced target planning into a multilayer GBAD framework to coordinate firepower allocation across defense layers. Karasakal [10] proposed differentiated WTA strategies for simultaneous and sequential attack patterns. Zhao et al. [7], Zhang et al. [8], and Xu et al. [9] further employed genetic algorithms, whale optimization, and beetle swarm optimization, respectively, to improve global search capability and convergence efficiency. As research gradually shifted from static WTA to Dynamic Weapon-Target Assignment (DWTA), adaptive dynamic programming [4], multi-objective optimization [34], and time-dependent allocation strategies [39] were introduced to address sequential decision-making problems. In addition, distributed optimization methods [36], [40], [41] have been proposed to support decentralized resource allocation under network-centric warfare.

Although these optimization-based methods have achieved satisfactory performance in predefined combat scenarios, they generally rely on handcrafted optimization strategies or predefined mathematical models. Consequently, they exhibit limited adaptability to highly dynamic battlefield environments and cannot effectively capture the evolving interactions among heterogeneous battlefield entities, which restricts their applicability to complex decentralized GBAD-WTA problems.

## 2.2 Learning-based WTA methods

The rapid development of deep reinforcement learning (DRL) has provided a promising alternative for solving dynamic WTA problems through adaptive sequential decision-making. By learning decision policies directly from battlefield interactions, DRL significantly improves real-time response capability compared with conventional optimization approaches. Representative studies include pointer-network-based end-to-end WTA [38], collaborative distributed reinforcement learning for multi-ship formations [17], and deep reinforcement learning frameworks incorporating prioritized experience replay and exploration mechanisms [20].

More recently, multi-agent reinforcement learning (MARL) has become an important research paradigm for decentralized WTA. Existing studies have introduced

multi-agent proximal policy optimization with attention mechanisms [21], curriculum learning strategies [22], and graph neural networks [42] to improve cooperative decision-making and battlefield representation. These methods demonstrate the potential of MARL in modeling decentralized interactions and learning adaptive policies for complex combat environments.

Nevertheless, several important challenges remain unresolved. First, most existing methods employ simplified state representations that inadequately characterize the dynamic spatio-temporal relationships among heterogeneous battlefield entities, including radar units, missile launchers, and incoming targets. Second, existing multi-objective reinforcement learning methods generally optimize multiple objectives through scalar reward aggregation, making it difficult to independently regulate different operational objectives during policy learning. These limitations reduce coordination efficiency and policy adaptability in realistic GBAD-WTA scenarios.

# 3 Problem formulation

## 3.1 Problem statement

The GBAD system operates through the tightly coupled coordination of heterogeneous nodes. Within an air defense interception mission, the effective allocation of radar, fire-power platforms, and interceptor resources is critical. As il-lustrated in Fig 2, the GBAD architecture can be stratified into four distinct layers: the sensor layer, firepower layer, interceptor layer, and the adversary??s threat layer. And GBAD-WTA extends beyond simple missile-to-target allocation; it necessitates synergistic collaboration across multiple nodes. Initially, the sensor layer must reliably detect and track emerging threats. Subsequently, the command system determines interception strategies based on real-time threat assessments??specifically, identifying the optimal firepower node for engagement. Finally, the intercepted target is neutralized as the launched interceptors, guided by the sensor layer, ensure precise terminal guidance. Consequently, the GBAD system constitutes a highly complex heterogeneous multi-agent system requiring intricate, cross-layer coordination. We consider a ground-based air defense system composed of $n$ SAM launchers and $m$ radar units, all of which can participate in coordinated air defense operations. SAM launchers operate in conjunction with radars to execute interception missions, with their respective detection and engagement ranges predefined. The system follows a standard interception pipeline consisting of detection?Ctracking?Cengagement?Cguidance?Cinterception, with missile velocity and minimum launch intervals explicitly specified.

Different weapon platforms carry varying numbers of missiles. Let $W$ denote the set of all missiles, and $w_i$ denote the number of missiles carried by SAM launcher $i$, if launcher $i$ is not equipped with missiles, then $w_i = 0$. A total of hostile aircraft approach the defended area from multiple directions. The entire air defense process is discretized into decision stages while the locations of SAM launchers and radar units are assumed to remain fixed. The defensive operation continues until the end of the engagement horizon, i.e., the completion of all stages. Critical symbols used in model-ling are summarized in Table 1.

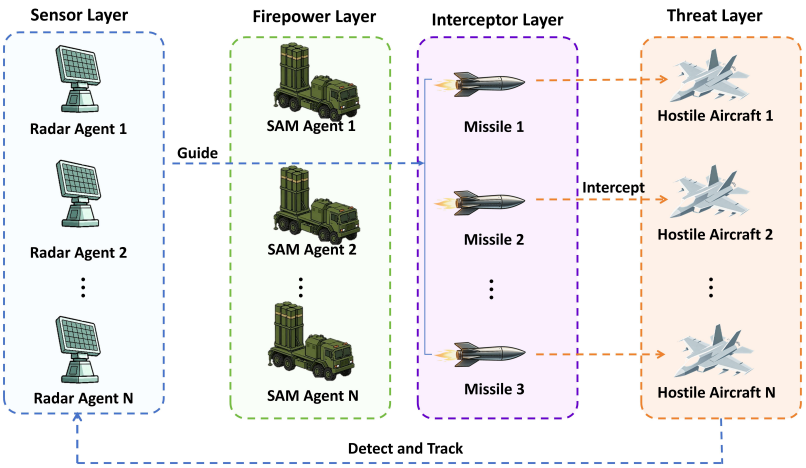



**Fig. 2** GBAD-WTA system in cooperative air defense scene.

**Table 1** Notation declaration in model

| Notation | Definition |
|---|---|
| $N$ | the set of surface-to-air missile launchers, $N = 1, 2, ..., n$ |
| $M$ | the set of radar systems, $M = 1, 2, ..., m$ |
| $W$ | the set of missiles, $W = 1, 2, ..., w$ |
| $w_i$ | the number of missiles carried by surface-to-air missile launcher $i$ |
| $K$ | the set of hostile targets, $K = 1, 2, ..., k$ |
| $T$ | the set of states in whole ground-based air defense process, $T = 1, 2, ..., t$ |
| $x_{ijk}(t)$ | the assignment of the missile from the SAM launcher $i$ to the hostile target $j$ at stage $t$ |
| $J(\cdot)$ | the objective function of GBAD-WTA decision making |
| $r$ | the engagement range of SAM launcher |
| $r'$ | the detection range of radar |
| $d$ | distance between target and SAM launcher or radar |
| $hp$ | the health value of target |

## 3.2 Objective function

In large-scale aerial swarm attack scenarios, the core objective of the GBAD-WTA problem is to determine an optimal target allocation strategy based on the available air defense resources, with the dual goal of minimizing the expected threat posed by surviving enemy targets while reducing overall air defense resource consumption [20]. Let the decision variable $x_{ijk}(t)$ denote the assignment of the missile $j$ from the SAM launcher $i$ to the hostile target at stage $t$:

$$x_{ijk}(t) = \begin{cases} 1, & \text{if } i \text{ and } j \text{ assigned to } k \text{ in stage } t \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

Due to weapon reliability limitations and the uncertainty associated with single-shot kill probability [23], let $g_{jk}$ denote the probability that a single missile $j$ launched by SAM launcher successfully destroys target $k$ and $h_i$ denote the weapon accuracy of SAM launcher $i$ .Then, at stage $t$, if target $k$ is engaged $x_{ijk}(t)$ times, Thus, in stage $t$ , the probability that target $k$ is not destroyed can be expressed as:

$$p_k(t) = 1 - \prod_{i \in N} \prod_{j \in W} \left(1 - h_i \cdot g_{ijk}(t)\right)^{x_{ijk}(t)} \tag{2}$$

in the following, we formulate the objective functions of the GBAD-WTA problem from two perspectives:

1) *Minimizing the expected survival threat of hostile targets.*Let denote the dynamic threat coefficient of hostile target at decision stage . Across all decision stages, the objective is to minimize the expected survival threat of hostile targets, which can be expressed as:

$$f_1(\boldsymbol{X}) = \sum_{k \in K} p_k(t)\eta_k(t) \tag{3}$$

Here, $\boldsymbol{X}$ denotes the matrix form of the decision variables ,the same notation applies hereafter.

2) *Minimizing platform ammunition consumption.* Minimizing ammunition consumption leads to lower operational costs. Since $x_{ijk}(t)$ is a binary variable, it naturally reflects whether a munition is used. Thus, the objective can be expressed directly in terms of the decision variables as:

$$f_2(\boldsymbol{X}) = \sum_{k \in K} \sum_{i \in N} \sum_{j \in W} x_{ijk}(t) \tag{4}$$

Based on (2), (3) and (4), the objective function of GBAD-WTA can be constructed as (5):

$$\begin{aligned} J(\boldsymbol{X}) = \min \sum_{t \in T} \{ [ \sum_{k \in K} (1 - \prod_{i \in N} \prod_{j \in W} (1 - h_i \cdot g_{ijk}(t))^{x_{ijk}(t)}) \times \eta_k(t)] \\ + [ \sum_{k \in K} \sum_{i \in N} \sum_{j \in W} x_{ijk}(t)] \} \end{aligned} \tag{5}$$

## 3.3 Constraints

The model we established includes the following four types of constraints:

1) *Firing frequency constraints.* Assuming that each SAM launcher can perform at most one firing action and launch a single missile per stage, the following constraints are imposed:

$$\sum_{j \in W} \sum_{k \in K} x_{ijk}(t) \leq 1, \quad \forall t \in T, \forall i \in N \tag{6}$$

Meanwhile, each firing action is subject to a cooldown constraint. If SAM launcher $i$ executes a launch at stage $t$, its cooldown is reset to a predefined firing cooldown

constant $c_{\text{fire}}$, and subsequently decreases during the stage update. Accordingly, the remaining cooldown time at the next stage can be expressed as:

$$c_i(t+1) = \begin{cases} c_{\text{fire}} - 1, & \text{if } \sum_{j \in W} \sum_{k \in K} x_{ijk}(t) = 1 \\ \max(0, c_i(t) - 1), & \text{otherwise} \end{cases} \tag{7}$$

It implies that the launcher must wait until the cooldown period is fully elapsed before it can fire again.

2)*Firepower resource constraints.* Each SAM launcher is equipped with a limited number of air defense missiles. Let $w_i$ denote the number of missiles available on launcher $i$. Once a given type of weapon is depleted, it can no longer be used for subsequent engagements. Accordingly, the corresponding constraints can be formulated as follows:

$$\sum_{t \in T} \sum_{k \in K} x_{ijk}(t) \leq w_i \tag{8}$$

3)*Engagement Feasibility Constraints.* Engagement feasibility constraints play a critical role in GBAD-WTA problems. In real-world combat scenarios, weapon-target engagements are subject to a variety of feasibility limitations, as illustrated in Fig. 3.Among these constraints, engagement range is the most influential factor. The deployment and utilization of air defense assets across different engagement ranges are particularly important in operational planning and execution [24]. To model these

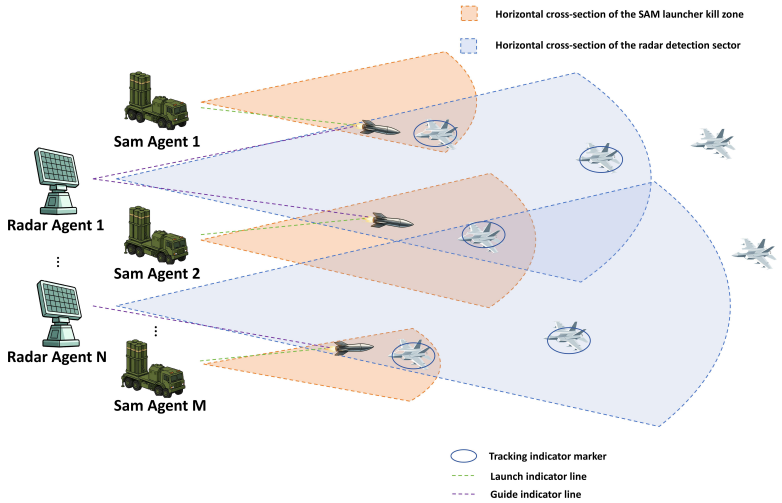



**Fig. 3** A simplified SAM launcher horizontal kill zone cross-section and radar detection cross-section, as well as inter-unit coordination.

constraints, Let $r_i$ denote the engagement range of SAM launcher $i$, and $r'_h$ denote the detection range of radar $h$. During stage $t$, let $d_{ikt}$ represent the distance between target $k$ and SAM launcher $i$, and $d_{hkt}$ represent the distance between target $k$ and

radar $h$. Furthermore, let $hp_k$ denote the health value of target $k$. For SAM launcher $i$ to engage target $k$, the following conditions must be satisfied:

I) The target is within the range of SAM launcher:

$$d_{ikt} \leq r_i, \quad \forall t \in T \tag{9}$$

II) The target lies within the detection and tracking range of at least one radar:

$$d_{hkt} \leq r'_h, \quad \forall h \in M, \forall t \in T \tag{10}$$

III) The target's health value remains positive, that is $hp_k > 0$.

IV) At least one radar has spare guidance capacity to support the launched missile.

Then, the engagement feasibility coefficient $f_{ijk}(t)$is introduced to indicate whether SAM launcher can employ weapon $j$ to engage target $k$ at stage $t$:

$$f_{ijk}(t) = \begin{cases} 1, & \text{if all conditions are met} \\ 0, & \text{otherwise} \end{cases} \tag{11}$$

The decision variable $x_{ijk}(t)$ must satisfy the engagement feasibility constraints, which can be formulated as follows:

$$x_{ijk} \leq f_{ijk}(t), \quad \forall i \in N, \forall j \in W, \forall k \in K, \forall t \in T \tag{12}$$

# 4 Proposed Method

## 4.1 Dec-POMDP

In this work, the GBAD-WTA problem in a cooperative heterogeneous multi-agent setting is formulated as a Dec-POMDP [16], characterized by the tuple$\zeta = \langle S, A, U, \Omega, O, P, R, n, \gamma \rangle$.

Here, $S$ denotes the global state space of the environment, and $A$ represents the set of $n$ heterogeneous agents involved in the task. In particular, the agent set can be partitioned into two types, $A_1$ and $A_2$, i.e., $A \equiv A_1 \cup A_2$,this corresponds to two types of agents in air defense systems: SAM launcher and radar. At each time step, each agent $k \in A$ selects an action $u_k \in U$, forming a joint action vector $\boldsymbol{u} = [u_1, ..., u_k]$. The joint action produces an immediate reward $r(s, u) \in R$ and induces a transition to the next state $s' \sim P(s' \mid s, u), P : S \times A \times S \to [0, 1]$ is the transition probability kernel [18].

Under partial observability, each agent receives an observation $o_k \in \Omega$ through the observation function $O(s, k)$, and the joint observation is $o \in \Omega \equiv \Omega^K$. At the end of each episode, each agent generates an action?Cobservation trajectory according to its learned stochastic policy , which is subsequently used to update the poli-cy. denotes the discount factor. The overall objective is to learn the optimal joint policy that

maximizes the expected discounted return [25]:

$$J(\pi) = \mathbb{E}_{\pi,p}\left[\sum_{t=0}^{\infty} \gamma^t r_t\right] \tag{13}$$

1) *Global State and Local Observation.* Let $S$ denote the global state space, and let $s_t \in S$ represent the global state $t$ at decision stage . The global state characterizes the overall battlefield situation in terms of target-related features. Specifically, it includes each target??s health value, dynamic threat coefficient, position, velocity, and detection status. Therefore, $s_t$ can be defined as $s_t = \langle HP, \psi, \boldsymbol{p}, \boldsymbol{v}, DE\rangle$.

Where $HP$ and $\psi$ are the remaining health and dynamic threat coefficient of target $1, 2, ..., k$,respectively. And $\boldsymbol{p} \in \mathbb{R}^2, \boldsymbol{v} \in \mathbb{R}^2$ denote their position and velocity, respectively. $DE \in 0, 1$ indicates whether the target has been detected. Notably, the dynamic threat coefficient is adaptively updated based on the target??s proximity to defended assets, reflecting time-varying operational priorities.

The system consists of heterogeneous agents, including radar agents $A_1$ and SAM launcher agents $A_2$, each with distinct observation structures and capabilities. Radar agents receives a local observation $o_t^{\text{rad}} = \langle o^{\text{target}}, r', cap_{track}, cap_{guide}, \boldsymbol{p}^{\text{rad}}\rangle$ While SAM launcher agents receives their local observation as $o_t^{\text{sam}} = \langle o^{\text{target}}, w, r, \eta, c, \boldsymbol{p}^{\text{sam}}\rangle$.

2) *Action space.* The action space is discrete and shared structurally across heterogeneous agents, but its semantics differ. For each agent $i$, the action is defined as $u_i \in 0, 1, ..., N_T$ where $u_i = k$ refers to select target $k$, and $u_i = N_T$ means to select a no-op action.

Specifically, for radar agents, selecting a target corresponds to assigning that target for tracking, whereas for SAM agents, it corresponds to launching a missile at the selected target. These actions are subject to feasibility constraints and cannot be executed arbitrarily. To enforce valid decision-making, an action masking mechanism is employed to re-strict the agent??s choices to feasible actions only. The mask design strictly follows the constraints defined in Section II.

3) *rewards.* For radar agent $i$ in $A_1$,the reward consists of four parts: target detection reward, target assignment reward, continuous tracking reward, and cooperative guidance reward. Specifically, when radar $i$ newly detects target $j$, it receives a reward proportional to the target threat; when it successfully assigns or locks target $j$ for tracking, it receives an additional assignment reward; while target $j$ remains under active track-ing, a small per-step tracking reward is accumulated; furthermore, if a SAM engagement supported by this radar results in a hit or kill, the radar receives an auxiliary cooperative reward, Thus, the radar reward can be written as

$$\begin{aligned} r_{i,t}^{\text{rad}} = & \underbrace{r_{i,t}^{\text{det}}}_{\text{detection reward}} + \underbrace{r_{i,t}^{\text{asg}}}_{\text{assignment reward}} + \underbrace{r_{i,t}^{\text{trk}}}_{\text{track reward}} \\ & + \underbrace{r_{i,t}^{\text{ah}}}_{\text{auxiliary hit reward}} + \underbrace{r_{i,t}^{\text{ak}}}_{\text{auxiliary kill reward}} - \underbrace{r_{i,t}^{\text{inv}}}_{\text{invalid action penalty}} \end{aligned} \tag{14}$$

For a SAM launcher agent $i$ in $A_2$, the reward encourages effective fire allocation while penalizing wasteful launches. When launcher $i$ fires a missile, it first incurs a fixed ammunition cost. If the missile successfully damages target $j$, the launcher receives a hit reward proportional to both the inflicted damage and the target threat. If the target is completely destroyed, an additional kill reward is granted. Therefore, the SAM reward is defined as

$$r_{i,t}^{\text{sam}} = -\underbrace{r_{i,t}^{\text{shot}}}_{\text{shot cost penalty}} + \underbrace{r_{i,t}^{\text{hit}}}_{\text{hit reward}} + \underbrace{r_{i,t}^{\text{kill}}}_{\text{kill reward}} - \underbrace{r_{i,t}^{\text{inv}}}_{\text{invalid action penalty}} \tag{15}$$

The meanings of the symbols used in Dec-POMDP sub-section are summarized in Table 2.

**Table 2** Notation declaration in DEC-POMDP

| Notation | Definition |
|---|---|
| $S$ | the global state space of the environment |
| $A$ | the set of heterogeneous agents involved in the task |
| $U$ | the action space of the agents |
| $\Omega$ | the global state space of the environment |
| $O$ | the observation space of the agents |
| $R$ | the reward set of agents receive from the interaction of the environment |
| $n$ | the number of agents |
| $\gamma$ | discount factor in discounted cumulative reward |
| $HP$ | the remaining health of target |
| $\psi$ | dynamic threat coefficient of target |
| $\boldsymbol{p}$ | the position vector of targets |
| $\boldsymbol{v}$ | the velocity vector of targets |
| $DE$ | the target detection status |
| $o^{\text{target}}$ | the state of the target locally observed by the target |
| $cap_{\text{track}}$ | radar tracking channel capacity |
| $cap_{\text{guide}}$ | radar guide channel capacity |
| $w$ | the amount of ammo available for the SAM launcher |
| $\eta$ | dynamic threat values of targets locally observed by the SAM launcher |
| $c$ | launch cooldown of the SAM launcher |
| $\boldsymbol{p}^{\text{rad}}$ | the position vector of radar agents |
| $\boldsymbol{p}^{\text{sam}}$ | the position vector of SAM launchers |

## 4.2 CTDE process

The proposed STG-MADAC framework follows the CTDE paradigm defined by the cooperative Dec-POMDP formulation in Section IV-A. During training, the actor-critic learner uses the global battlefield state, the complete joint trajectory, and role-aware reward signals to estimate coordinated value targets and policy gradients. During execution, each radar or SAM launcher acts only from its local observation and feasible-action mask. As shown in Fig.4, STG-MADAC therefore organizes the learning process around a centralized critic, heterogenous decentralized actors,

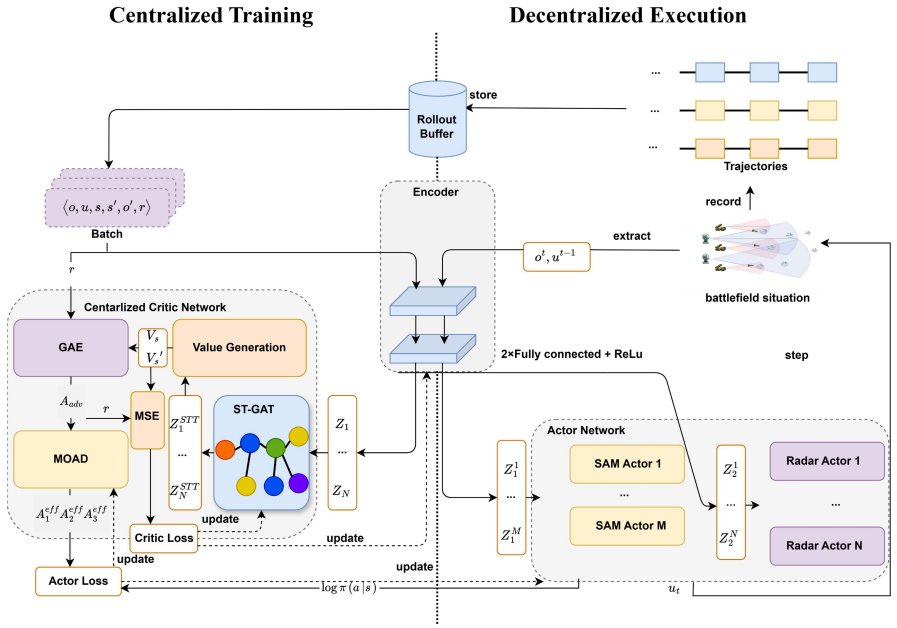



**Fig. 4** Schematic illustration of STG-MADAC with a centralized training and decentralized execution architecture for GBAD-WTA.

a spatio-temporal graph situation encoder, and an adaptive multi-objective advantage calculation. The detailed structure of graph attention networks and advantage decomposers will be explained in Section IV-C, and the complete training procedure is summarized in Section IV-D.

1) *Centralized training.* At decision stage $t$, the environ-ment provides the global state $s^t$, local observations $o_{i\,i=1}^{t\,N}$, joint action $u^t = (u_1^t, ..., u_N^t)$, and multi-objective reward vector $\boldsymbol{r}_i^t = [r_{i,1}^t, ..., r_{i,K}^t]$.

These information forms the collected trajectories and are first fed into each agent??s individual feature encoder, which consists of a two-layer MLP with linear transformations followed by ReLU activations. The encoder produces the individual feature representation $Z_t$, these features are then passed to the spatio-temporal graph-attention situational encoder which generates graph situational representations for centralized value estimation.

Based on the encoded spatio-temporal situation, represented as $Z_t^{STT}$, The centralized critic estimates objective-specific value functions and computes temporal-difference (TD) errors for each objective. These TD errors are used to construct generalized advantage estimates. An adaptive mul-ti-objective advantage decomposition module transforms the resulting advantages into effective advantages for policy optimization. The critic is updated by minimizing TD errors via backpropagation, while decentralized actors are optimized using policy gradient methods.

2) *Decentralized execution.* During decentralized execution, the centralized critic, spatio-temporal graph encoder, and adaptive multi-objective advantage decomposition mod-ule are removed. Each agent independently selects actions based only on its

local observation and feasible-action mask $u_i^t \sim \pi_{\theta_i}(u_i^t \mid o_i^t, m_i^t)$. At this stage, the trajectories collected from the environment are first processed by each agent??s individual feature encoder, which outputs the individual feature representation $Z_t$. The key difference in STG-MADAC is that each agent is equipped with an independent actor network. The feature $Z_t$ is then distributed to each actor, which selects actions based on its corresponding local features and network parameters.

After all agents execute their actions, the environment will be updated according to joint action $u^t$ and multi-objective reward signals are recorded for the next training update. In this way, STG-MADAC exploits global graph-structured battlefield information during training while pre-serving decentralized and deployable decision-making during execution.

## 4.3 Network design

1) *Spatio-temporal graph attentional situation encoder.* To improve the situational representation capability of the centralized critic, STG-MADAC introduces a spatio-temporal graph attentional situation encoder before value estimation. As Fig 5 shows, it transforms the global battlefield state into a unified graph representation, where spatial dependencies among targets, radar systems, and SAM launchers are explicitly modeled via relation-aware graph attention. Meanwhile, temporal evolution is implicitly captured through recurrent state updates in sequential decision-making. Formally,

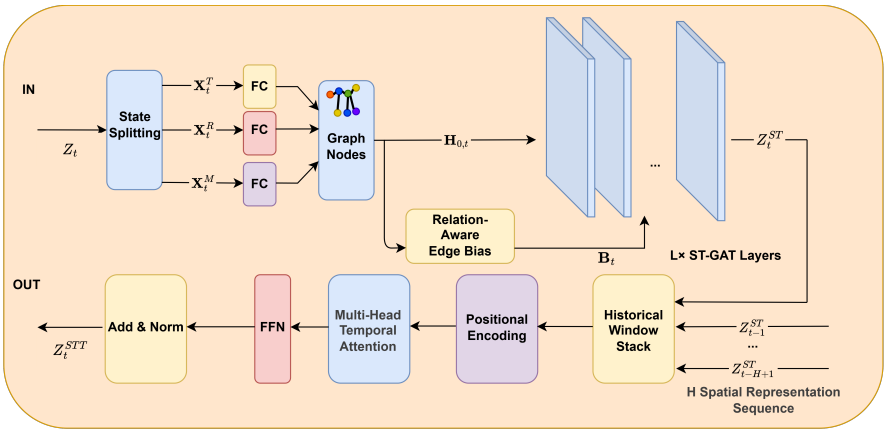



**Fig. 5** the network structure of spatio-temporal graph attentional situation encoder.

at each time step $t$, the centralized individual feature representation $Z_t$ is spitted to vector $[\boldsymbol{X}_t^T, \boldsymbol{X}_t^R, \boldsymbol{X}_t^M]$, where $\boldsymbol{X}_t^T$,$\boldsymbol{X}_t^R$, and$\boldsymbol{X}_t^M$] denote feature sets of target, radar, and missile launcher nodes, respectively. Then each entity type is first encoded into a unified latent space using type-specific linear projections and forms the initial node embed-ding of the battlefield graph $\boldsymbol{H}_0 = [\boldsymbol{H}_0^T, \boldsymbol{H}_0^R, \boldsymbol{H}_0^M]$.

To capture spatial interactions, we construct a fully connected graph with a relation-aware edge bias matrix $\boldsymbol{B}_t$, which encodes physically meaningful relationships such as distance,threat level, relative velocity, engagement range, and operational readiness.

At stage $t$,graph nodes $\boldsymbol{H}_0^t$ is send to ST-GAT layers, flows through all layers in sequence and in specific ST-GAT layer $L$, given node embeddings $\boldsymbol{H}_{l-1}^t$, we compute query, key, and value projections [30]:

$$\begin{cases} Q = \boldsymbol{H}_{l-1,t} W_Q \\ K = \boldsymbol{H}_{l-1,t} W_K \\ V = \boldsymbol{H}_{l-1,t} W_V \end{cases} \tag{16}$$

The attention score then is modified by incorporating the relation-aware bias, and multi-head outputs are concatenated and linearly projected by:

$$\begin{aligned} \text{head}_i &= \text{softmax}\left(\frac{QK^T}{\sqrt{d}} + B_t\right) V_i \\ \tilde{\boldsymbol{H}}_{l,t} &= \text{concat}\big(\text{head}_1, \ldots, \text{head}_h\big) W_O \end{aligned} \tag{17}$$

Then, a FFN, consisting of two linear layers and one Ge-LU layer, is applied after attention aggregation. The layer adopts residual connections and layer normalization to form the output $\boldsymbol{H}_l$:

$$\boldsymbol{H}_{l,t} = \text{LN}\left(\boldsymbol{H}_{l-1,t} + \text{FFN}\left(\text{LN}\left(\boldsymbol{H}_{l-1,t} + \tilde{\boldsymbol{H}}_{l-1,t}\right)\right)\right) \tag{18}$$

After graph reasoning, node embeddings are split according to entity type $\boldsymbol{H}_0^T, \boldsymbol{H}_0^R, \boldsymbol{H}_0^M$, each type is projected back to its original feature space with residual refinement:

$$\hat{\boldsymbol{X}}_t^\tau = \boldsymbol{X}_t^\tau + \varphi_\tau\left(\boldsymbol{H}_l^\tau\right), \quad \tau \in \{T, R, M\} \tag{19}$$

The enhanced centralized state is then reconstructed through flatten operation to form $Z_t^S T$, then, together with the former spatial representation sequence, the proposed spatio-temporal encoder integrates heterogeneous spatial reason-ing and temporal information to produce a structured and semantically enriched battlefield representation $Z_t^{STT}$ for multi-agent decision making.

2) *Adaptive multi-objective advantage decomposer.* Inspired by the multi-agent decomposition theorem [26], which shows that the joint advantage of a cooperative multi-agent system can be decomposed into a sum of sequential contributions along a given agent ordering, we extend this principle beyond agent-wise decomposition and propose an objective-aware advantage decomposition framework. Instead of decomposition advantages across agents, we decompose the scalar advantage into $K$ objective components:$A_i(o,u) \to A_{i,k}(o,u)_{k=1}^K$, where each component correspond to a distinct optimization objective, such as threat suppression, resource efficiency, or coordination quality.

To reconstruct a task-relevant advantage signal, our learned multi-objective advantage decomposition (MOAD) module adaptively aggregates objective-wise advantages:

$$A_i^{\text{eff}} = \sum_{k=1}^{K} w_k(s_t) A_i \cdot g_k(s_t), \quad \text{s.t.} \quad \sum_{k=1}^{K} w_k = 1 \tag{20}$$

where $w_k(s_t)$ is a state-dependent weighting function learned by a neural network and

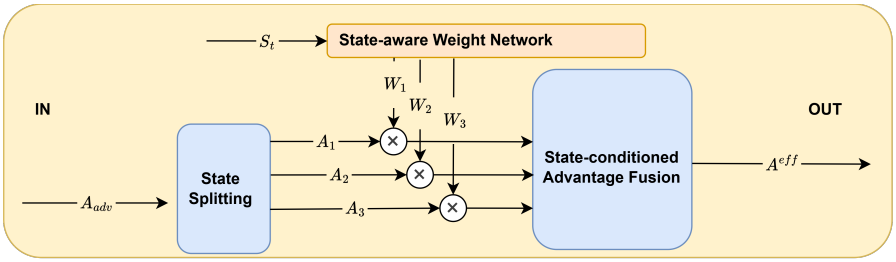



**Fig. 6** multi-objective advantage decomposer structure.

$g_k(\cdot)$ captures nonlinear interactions between the global state and each optimization objective. The structure of MOAD is exhibited in Fig 6.

## 4.4 Algorithm procedure

This subsection connects shows the detailed algorithm training procedure of STG-MADAC. For a trajectory segment collected from the GBAD-WTA environment, the rollout batch can be denoted as

$$\tau_t = \langle s^t, (o_i^t)_{i=1}^N, u^t, (r_i^t)_{i=1}^N, s^{t+1}, (o_i^t)_{i=1}^N o_i^{t+1\,N}{}_{i=1} \rangle \tag{21}$$

The individual features are organized as a heterogeneous battlefield graph and encoded by the ST-GAT module. The centralized critic estimates objective-specific values based on the representation , and for objective $V_{\phi,k}\left(S_t, \boldsymbol{Z}_t^{STT}\right), \quad k = 1, \ldots, K$, the temporal-difference residual and advantage are computed as

$$\begin{aligned} \delta_{i,k}^t &= r_{i,k}^t + \gamma V_{\phi,k}\left(s_{t+1}, z_{t+1}^{STT}\right) - V_{\phi,k}\left(s_t, z_t^{STT}\right) \\ A_{i,k}^t &= \delta_{i,k}^t + \sum_{l=1}^{T-t} (\gamma\lambda)^l \delta_{i,k}^{t+l} \end{aligned} \tag{22}$$

MOAD aggregates these objective-wise advantages into the effective advantage used by policy improvement, the actor network is then updated in a clipped actor-critic form using this effective advantage [43], meanwhile??the centralized critic and the representation module are optimized through objective-specific value losses.

**Algorithm 1** STGMADAC Algorithm

**Require:** Environment $\mathcal{E}$, agent groups $\mathcal{G}$, actor $\{\pi_{\theta_i}\}_{i=1}^{N}$, critic $V_\phi$, ST-GAT encoder $f_\psi$, MOAD module $F_\omega$, $\gamma, \lambda, \epsilon$

**Ensure:** Optimized policy parameters $\theta^*$

Initialize $\theta, \phi, \psi, \omega$
**while** not converged **do**
  Collect trajectories $\tau = \{(s_t, o_t, a_t, r_t)\}_{t=0}^{T}$
  Obtain graph-aware state embedding: $z_t = f_\psi(s_t)$
  Compute value estimation: $V_t = V_\phi(z_t)$
  Calculate GAE advantage: $A_t = \text{GAE}(r_t, V_t, \gamma, \lambda)$
  Generate adaptive advantage: $A_t^{eff} = F_\omega(s_t, A_t)$
  **for** $g \in \mathcal{G}$ **do**
    Generate update order $\mathcal{O} = (i_1, \ldots, i_N)$ and set $F_0 = 1$
    **for** $j = 1, \ldots, N$ **do**
      Select agent $i = i_j$
      Compute PPO ratio:

$$r_i = \frac{\pi_{\theta_i}(a_i|o_i)}{\pi_{\theta_i^{old}}(a_i|o_i)}$$

      Update actor using:

$$\mathcal{L}_i = -F_{j-1} \min(r_i A_t^{eff}, clip(r_i, 1-\epsilon, 1+\epsilon) A_t^{eff})$$

      $\theta_i \leftarrow \theta_i - \alpha \nabla_{\theta_i} \mathcal{L}_i$
      Update sequential factor:

$$F_j = F_{j-1} \frac{\pi_{\theta_i^{new}}}{\pi_{\theta_i^{old}}}$$

    **end for**
  **end for**
  Update critic: $\phi \leftarrow \phi - \alpha_v \nabla_\phi \mathcal{L}_V$
  Update MOAD: $\omega \leftarrow \omega - \alpha_m \nabla_\omega \mathcal{L}_{MOAD}$
**end while**
**return** $\theta^*$

# 5 Computational experiments

## 5.1 Testbed,settings and configurations

All experiments were conducted on a same computational testbed.The experimental platform and software environment is shown in Table 3.All methods were implemented and evaluated under identical experimental settings, with detailed configurations provided in Table 4.

## 5.2 Metrics

To evaluate the effectiveness of STG-MADAC, three types of experiments were conducted: ablation studies to examine the contribution of key components, interpretability analyses to investigate the decision-making behaviors of the proposed

**Table 3** Experimental Platform and Software Environment

| Category | Configuration |
|---|---|
| Operating System | Windows 11 |
| Processor | Intel Core i5-14490F |
| Memory | 32 GB RAM |
| GPU | NVIDIA RTX 5070 |
| Development Language | Python 3.9 |
| Simulation Framework | Gymnasium + PettingZoo |
| Simulation System | GBAD-WTA simulation environment |

**Table 4** Simulation configuration parameters

| Items | Configuration | Items | Configuration |
|---|---|---|---|
| Scale of the battlefield | $100 \times 100$ | Discount factor$\gamma$ | 0.99 |
| Max number of decision steps | 50 | GAE coefficient$\lambda$ | 0.95 |
| Ammo capacity per weapon | $[12, 15]$ | PPO clip rate | 0.2 |
| Weapon accuracy | $[0.8, 0.9]$ | Entropy coefficient | 0.01 |
| Number of cooling steps during launch | $[3, 5]$ | Learning rate | $1 \times 10^{-4}$ |
| Target health value | $[0.7, 1.0]$ | Frames collected per batch | 6144 |
| Initial threat value of conventional targets | $[0.35, 1.65]$ | Minibatch size | 1024 |
| Initial threat value of high-threat targets | $[3.2, 4.6]$ | Number of training rounds per batch | 2 |
| Hit rewards | 5.5 | Number of parallel environments | 10 |
| Ammo consumption penalty | 0.28 | Evaluation interval | $120k$ frames |
| Action space | Discrete | Evaluation episodes | 10 |

framework, and comparative experiments against representative MARL baselines. In addition, perturbation experiments were performed to evaluate the robustness and adaptability of STG-MADAC under dynamic environmental variations.

Four metrics were used for performance evaluation: cumulative reward, number of intercepted targets, objective function value, and total threat value of surviving targets. Higher cumulative reward and interception number indicate better performance, while lower objective function value and residual threat value represent improved operational effectiveness.

## 5.3 Ablation experiments

Ablation experiments were conducted to evaluate the contribution of the spatial attention, temporal attention, and multi-objective advantage decomposition (MOAD) modules. The configurations of STG-MADAC network is shown in Table 5.Each variant was trained and evaluated under the same settings, and the results are summarized in Table 6 and Fig. 8. The complete STG-MADAC achieves the best overall performance, demonstrating the effectiveness of combining spatio-temporal representation learning and multi-objective optimization. Removing MOAD leads to the largest performance degradation, indicating its importance in balancing conflicting objectives.




**Fig. 7** GBAD-WTA simulation scene.

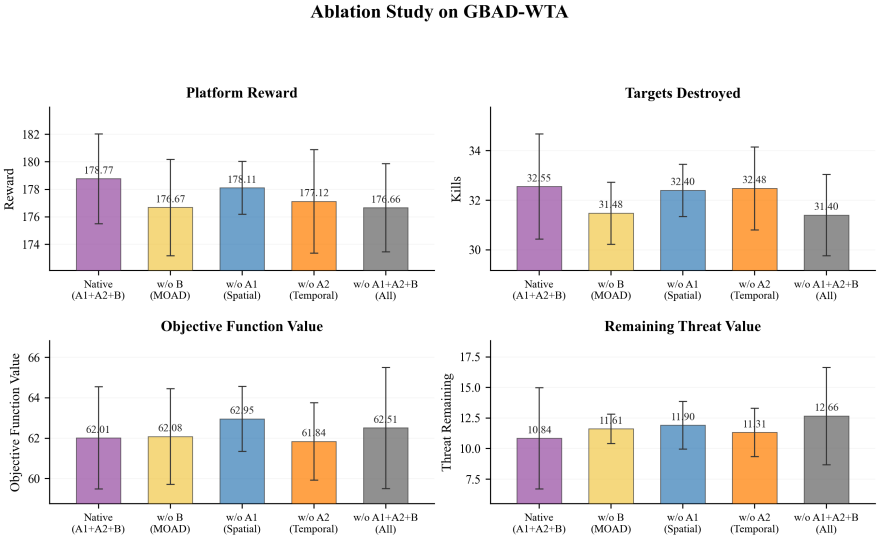



**Fig. 8** STG-MADAC ablation experiment results. The column heights represent the average of five random seeds. A1, A2, and B denote the spatial map attention, temporal attention, and multi-object advantage decomposition modules, respectively; w/o indicates without.

The removal of spatial or temporal attention also decreases performance, confirming that both spatial relationships and historical information contribute to effective sequential decision-making.

**Table 5** STGMADAC network configuration settings

| Items | Configuration | Items | Configuration |
|---|---|---|---|
| GAT hidden dim | 128 | Temporal layers | 1 |
| GAT heads | 4 | Target feature dim | 7 |
| GAT layers | 2 | MOAD objectives | 3 |
| Temporal hidden dim | 128 | MOAD temperature | 0.35 |
| Temporal heads | 4 | MOAD modulation strength | 0.65 |

**Table 6** Performance comparison of STG-MADAC and ablation variants

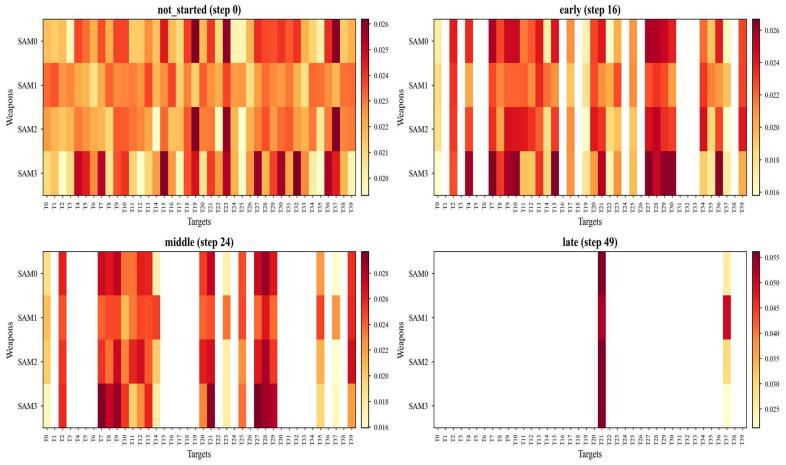
| Metric | STG-MADAC | Move MOAD | Move Spatial_attention | Move Temporal_attention | Move All |
|---|---|---|---|---|---|
| Reward mean | $178.77 \pm 2.62$ | $176.67 \pm 2.82$ | $178.11 \pm 1.55$ | $177.12 \pm 3.03$ | $176.66 \pm 2.59$ |
| Kills mean | $32.55 \pm 1.71$ | $31.48 \pm 1.01$ | $32.40 \pm 0.85$ | $32.48 \pm 1.35$ | $31.40 \pm 1.32$ |
| Objective function value | $62.01 \pm 2.04$ | $62.08 \pm 1.91$ | $62.95 \pm 1.30$ | $61.84 \pm 1.54$ | $62.51 \pm 2.41$ |
| Threat remaining mean | $10.84 \pm 3.35$ | $11.61 \pm 0.97$ | $11.90 \pm 1.57$ | $11.31 \pm 1.59$ | $12.66 \pm 3.21$ |

## 5.4 Interpretability analysis

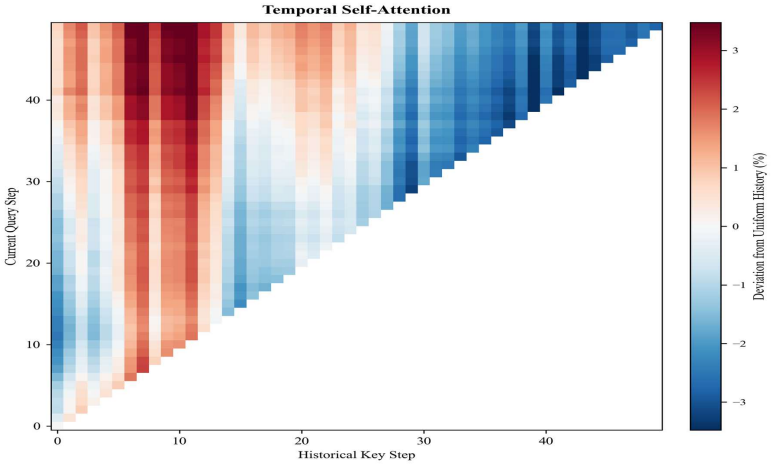
In this experiment, the attention weights from the graph attention network (GAT) layer and the self-attention weights from the temporal Transformer module were extracted to investigate how spatial and temporal attention mechanisms influence agent decision-making. By analyzing the distribution of attention weights, the contribution of different battle-field entities and historical states to the decision process can be further revealed. The results are presented under an experimental con-figuration with a ratio of 1:2:20 among launch platforms, radar units, and target nodes.

In terms of spatial attention, Fig. 9(a) shows that different SAMs gradually form consistent attention patterns toward critical targets. During the engagement process, multiple weapon nodes converge on high-threat targets (e.g., T7, T27, and T28), indicating that spatial attention enables collaborative target prioritization. In the final stage, when only a few targets remain, SAMs exhibit strong consensus on the primary threat, demonstrating the ability of spatial attention to facilitate coordinated decision-making among heterogeneous agents.

Regarding temporal attention, Fig. 9(b) indicates that the model does not simply focus on the latest states but selectively attends to important historical information. Several early historical intervals maintain strong influence across different query steps, suggesting that the agents can capture long-term temporal dependencies for sequential decision-making. To reveal the extent to which multi-objective dominance decomposition influences the final allocation scheme, by extracting decision paths and key edges, we investigated the overlap between key edges and decision schemes, introducing the Jensen-Shannon divergence metric [44]. We measure the difference in probability distributions for candidate targets between the two approaches with and without the MOAD mechanism. For the same SAM, let the action distributions of the Native strategy and the strategy with MOAD removed be and , then in this experiment, JS




(a)




(b)

**Fig. 9** Analysis of temporal and spatial attention weights. (a) shows the spatial attention weights for four typical phases in the selected evaluation rounds; the horizontal axis represents the target node, the vertical axis represents the SAM weapon node, and the colors indicate the multi-head average attention weights of the SAM launcher query pointing to the target key in the final layer of ST-GAT; (b) shows the self-attention matrix of the temporal Transformer for this evaluation round. The x-axis represents historical key steps, and the y-axis represents the current query step.

divergence is Defined as

$$
\begin{aligned}
D_{JS}(P\|Q) &= \frac{1}{2}D_{KL}(P\|M) + \frac{1}{2}D_{KL}(Q\|M), \\
M &= \frac{1}{2}(P+Q).
\end{aligned}
\tag{23}
$$

Its range is $[0, \ln 2]$. Generally speaking, the larger the JS divergence value, the more pronounced the difference in the probability distributions of the two models for the candidate targets. The experimental results are shown in Fig. 10.

The results show that, based on changes in the weights of the three MOAD target categories, the weight for threat reduction was significantly higher overall than those for resource conservation and radar-SAM coordination during the initial phase, indicating that when the number of enemy targets was high and the remaining threat level was significant, MOAD primarily emphasized suppressing surviving threats, after step

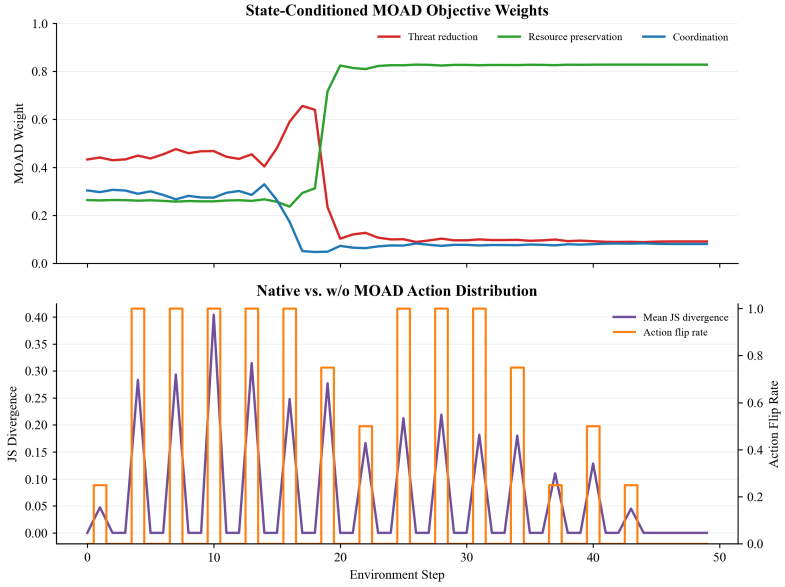



**Fig. 10** MOAD target wets generated by native STG-MADAC over the course of an evaluation round as the environ-ment progresses through steps. The Figure compares the differences in action distributions between Native STG-MADAC and a separately trained strategy without MOAD under the same observations and action masks; the purple curve represents the average JS divergence of the 4 SAM action distributions, and the orange stepped line indicates the proportion of SAMs where deterministic actions were flipped.

20, the resource conservation weight increased rapidly, indicating that the agent gradually prioritized resource-efficient strategies as ammunition consumption ac-cumulated. Compared with the scenario without MOAD, STG-MADAC exhibited action reversals in 15 of 50 steps, leading to 45 SAM decision changes. Step 10 showed the highest JS divergence with a 100% action reversal rate, indicating that MOAD can alter both action preferences and final allocation decisions. These results demonstrate that MOAD continuously improves decision-making by adaptively balancing multiple objectives.

Furthermore, a decision-tree-based policy distillation approach [45-47]is employed to approximate the decisions generated by the proposed deep reinforcement learning model. By fitting the model outputs with an interpretable surrogate decision tree,

approximate decision rules are extracted and visualized to illustrate how the policies learned by the neural network are transformed into specific engagement decisions. Furthermore, decision scenario diagrams are generated to provide insights into the cooperative behaviours learned by heterogeneous agents.

Specifically, starting from a target detected and tracked by a radar unit at a given time step, we investigate whether the target is engaged by a surface-to-air missile (SAM) unit within the following three decision steps and define this pro-cess as a radar-to-SAM rapid engagement event. Three key variables, including radar channel capacity, available ammunition, and target residual threat level, are selected to construct the decision-tree approximation rules. The resulting decision tree and corresponding analysis are presented in Fig. 11. The extracted decision tree with a depth

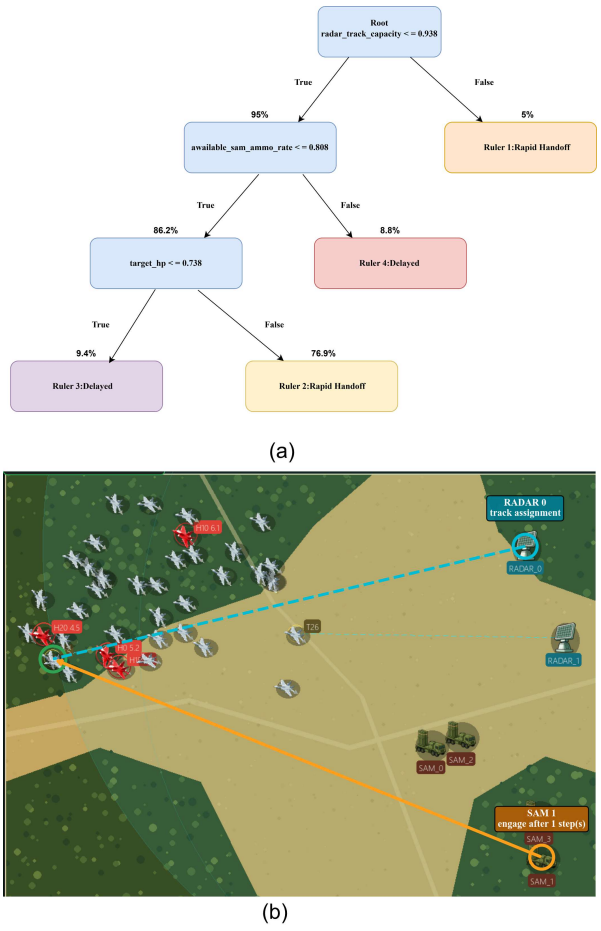



**Fig. 11** STG-MADAC Collaborative rules for heterogeneous agents and a typical decision-making backtracking process. The left side shows the rapid engagement rules for agents extracted from decision trees; the right side displays a typical collabo-rative process rendered from a simulation scenario.

of 3 effectively reveals the cooperative behaviours among heterogeneous agents. When radar resources are sufficient, agents tend to follow Rule 1 for target engagement. As radar resource utilization increases, Rule 2 becomes dominant, leading agents to prioritize high-threat targets, which accounts for 76.9% of the samples. Rule 3 indicates that damaged targets are often as-signed lower engagement priority, while Rule 4 shows that sufficient ammunition availability may lead to delayed engagement in a small number of cases. These extracted rules demonstrate that STG-MADAC learns adaptive coordination strategies consistent with practical engagement principles.

### 5.5 Comparative experiment

STG-MADAC was compared with representative MARL algorithms, including MAPPO [48], QMIX [49], QPLEX [50], and IPPO [51]. The quantitative results under different experimental settings are summarized in Table 7, and the learning curves for the 1:2:20 scenario are presented in Fig. 12. STG-MADAC achieves better overall performance than the baseline methods across the evaluation metrics, with higher cumulative rewards and lower objective values and residual threat levels. This demonstrates that the proposed framework can effectively coordinate heterogeneous agents and improve the trade-off between resource utilization and operational effectiveness. The results verify the effectiveness of combining spatio-temporal representation learning with multi-objective decision optimization for dynamic resource allocation problems.

**Table 7** Performance comparison of different multi-agent reinforcement learning algorithms

| **Metric** | **STG-MADAC** | **MAPPO** | **QMIX** | **QPLEX** | **IPPO** |
|---|---|---|---|---|---|
| Reward mean | $178.77 \pm 2.62$ | $176.67 \pm 2.82$ | $178.11 \pm 1.55$ | $177.12 \pm 3.03$ | $177.12 \pm 3.03$ |
| Kills mean | $32.55 \pm 1.71$ | $31.48 \pm 1.01$ | $32.40 \pm 0.85$ | $32.48 \pm 1.35$ | $32.48 \pm 1.35$ |
| Objective function value | $62.00 \pm 1.34$ | $86.77 \pm 1.39$ | $84.72 \pm 3.47$ | $85.93 \pm 1.21$ | $85.73 \pm 0.97$ |
| Threat remaining mean | $10.84 \pm 3.35$ | $11.61 \pm 0.97$ | $11.90 \pm 1.57$ | $11.31 \pm 1.59$ | $11.31 \pm 1.59$ |

## 6 Conclusion

This paper investigated a decentralized weapon?Ctarget assignment problem in ground-based air defense systems involving heterogeneous resources and dynamic operational conditions. A multi-agent reinforcement learning framework, STG-MADAC, was developed by integrating spatio-temporal representation learning and multi-objective advantage decomposition to improve cooperative resource allocation and sequential decision-making. Computational experiments demonstrated that the proposed approach achieves competitive performance compared with representative MARL methods and provides interpretable insights into agent coordination and target prioritization. The proposed framework offers a data-driven decision-support approach for complex operational systems. Future work will extend the model by incorporating

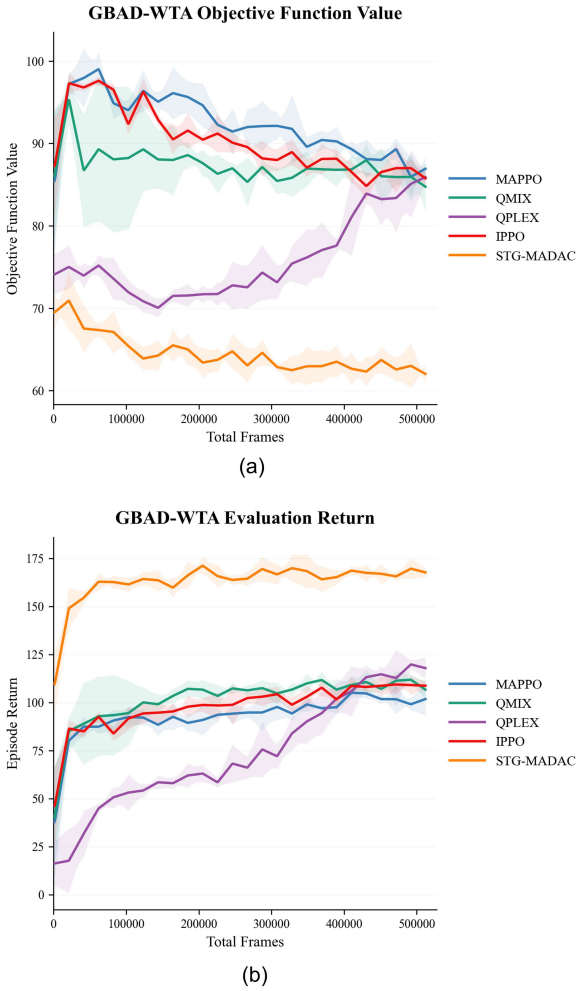



**Fig. 12** Evaluation return and objective function value in Experimental Results.

additional uncertainties and heterogeneous entities, such as electronic warfare effects and swarm-based defense scenarios.

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- The authors declare that they have no conflict of interest.
- Ethics Approval and Consent to Participate:Not applicable.

- Consent for Publication:Not applicable.
- The data that support the findings of this study are available from the corresponding author upon reasonable request.
- Materials Availability:Not applicable.
- The source code of the proposed GBAD-WTA simulation platform is available from the corresponding author upon reasonable request and will be made publicly available after publication.
- Author contribution:
  Haoyan Yao: Conceptualization, Methodology, Algorithm design, Software implementation, Experiments, Writing??original draft.
  Changan Shang: Experimental validation, Data analysis, Visualization.
  Wenzhe Zhang: Methodology, Results analysis, Writing??review and editing.
  Panrong Wang: Resources, Project administration, Supervision.
  Shengjie Xu: Conceptualization, Funding acquisition, Supervision, Writing??review and editing.

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